

5 **(short)** A process involving a perfect gas is shown in Fig. 5, whereby two gas streams are mixed, and heat is extracted, such that $\dot{Q}_{3-4} = -1.55$ MW. At inlet 1, the mass flow rate is $\dot{m}_1 = 1 \text{ kg s}^{-1}$ and the temperature is $T_1 = 1000 \text{ K}$. At inlet 2, the mass flow rate is $\dot{m}_2 = 3 \text{ kg s}^{-1}$ and the temperature is $T_2 = 2000 \text{ K}$. The pressure changes are negligible, and the gas has a constant specific heat capacity at constant pressure of $c_p = 350 \text{ J kg}^{-1} \text{ K}^{-1}$.

(a) Neglecting changes in potential and kinetic energy of the gases, calculate the temperatures after mixing T_3 , and after heat extraction T_4 . [5]

(b) Calculate the total rate of entropy generation for the mixing process, $\dot{S}_{1\&2-3}$. [5]

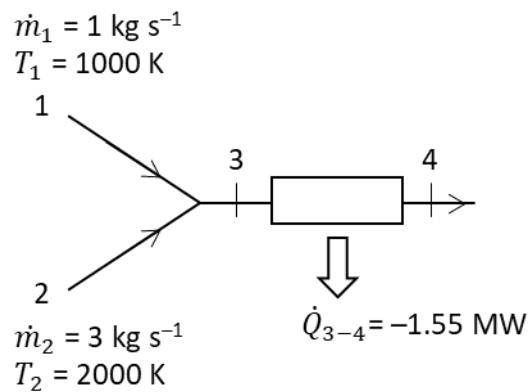


Fig. 5